

Concept Question 4.1 The Sagittarius Galaxy

Despite its proximity to the Milky Way, the dwarf galaxy in Sagittarius (Sagittarius Dwarf Spheroidal Galaxy) was not discovered until 1994. Can you think of a reason why it was not discovered earlier?

Hint: Consider what else is in the constellation.

Problem 4.2 The Plummer model

The Plummer model is a spherical density model with a “soft edge” that can be used to describe clusters. The Plummer gravitational potential and density profile are given by

$$\Phi(r) = -\frac{GM}{\sqrt{r^2 + b^2}}, \quad (1)$$

$$\rho(r) = \frac{3M}{4\pi} \frac{b^2}{(r^2 + b^2)^{5/2}}. \quad (2)$$

- a) Calculate and plot the mass profile $M(r)$ and the circular speed profile $v_c(r)$, as a function of radius. To solve the integrals you can use tools like <https://www.wolframalpha.com/calculators/integral-calculator/>
- b) Calculate the mean density for $r \leq b$ and compare it with $\rho(0)$.
- c) Calculate the total mass from $M(r)$, the potential energy E_{pot} , and the gravitational radius r_g . Compare these with the total mass and potential energy obtained for a homogeneous sphere of radius $r = b$ and mass M .